**1** **(a)** Figure 1.1 shows a speed–time graph for a toy aircraft.

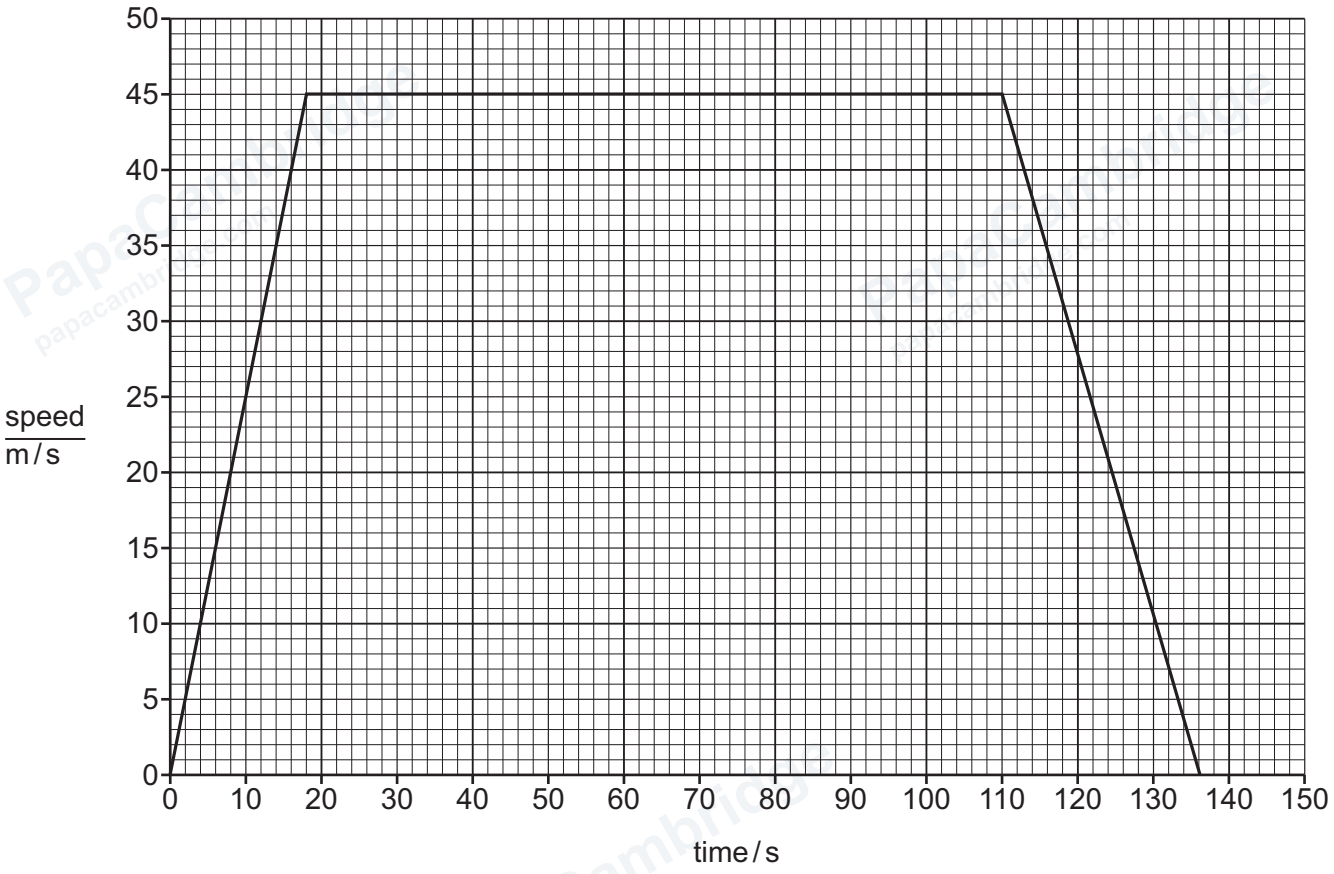


**Figure 1.1**

**(i)** Describe the motion between times:

18 s and 110 s .....

110 s and 136 s. .....

[2]

**(ii)** Calculate the distance travelled between time = 0 and time = 110 s.

distance = ..... m [3]

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**(iii)** Calculate the acceleration between time = 0 and time = 18 s.

acceleration = ..... m/s$^2$ [2]

**(b)** A small aircraft travels at a constant speed of 60 m/s. The mass of the small aircraft is 820 kg.

Calculate the energy $E_k$ in the kinetic store of the aircraft.

$E_k$ = ..... J [2]

[Total: 9]

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